

- 3 (a) *either* change in volume = $(1.69 - 1.00 \times 10^{-3})$
or liquid volume $\ll$ volume of vapour M1
 work done = $1.01 \times 10^5 \times 1.69 = 1.71 \times 10^5$ (J) A1 [2]
- (b) (i) 1. heating of system/thermal energy supplied to the system B1 [1]
2. work done on the system B1 [1]
- (ii) $\Delta U = (2.26 \times 10^6) - (1.71 \times 10^5)$ C1
 $= 2.09 \times 10^6$ J (3 s.f. needed) A1 [2]